20. Implement the Dilation Morphological Operation Using OpenCV

Code :

```
import cv2
import numpy as np

# Read the input image
img = cv2.imread(r"C:\Users\ROHIT\Downloads\download (1).jpeg")

if img is None:
    print("Error: Image not found")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Convert to binary image
    _, binary = cv2.threshold(
        gray, 127, 255, cv2.THRESH_BINARY
    )

    # Create a 3x3 structuring element
    kernel = np.ones((3, 3), np.uint8)

    # Apply dilation
    dilation = cv2.dilate(binary, kernel, iterations=1)

    cv2.imshow("Original Image", img)
    cv2.imshow("Binary Image", binary)
    cv2.imshow("Dilation Image", dilation)
```

```
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

output:

